



Global Child Mortality Analysis

A Full-Pipeline Data Analysis Project



Presented by:
Treasure Okafor



THE PROBLEM & OBJECTIVE

- **The Problem:** Global child mortality is declining, but is it just because of money (GDP)?
- **The Goal:** To investigate the correlation between wealth and health and identify "Success Story" countries that outperform their economic status.



KEY METRICS



CHILD MORTALITY RATE

Measures the deaths of children under five per 1,000 live births



GDP PER CAPITA

A country's total economic output (GDP) divided by its population



REGIONAL DATA

Geographically specific statistics



POPULATION

The total number of people in a specific area

DATA PREPARATION



Cleaning (Python)

Removed regional aggregates (like "World") to prevent overpopulation error.

```
[ ] # 1. Handle Missing Values
# Drop rows where critical variables (CMR or GDP) are missing
df = df.dropna(subset=['Child mortality rate', 'GDP per capita'])

[ ] # 2. Fix the Columns Names
# Let's shorten them so they are easier to type
df = df.rename(columns={
    'Child mortality rate': 'CMR',
    'GDP per capita': 'GDP',
    'Population (historical)': 'Population',
    'World regions according to OWID': 'Region'
})

print('Column names fixed')

Column names fixed

[ ] # 3. Handle the "Continent vs Country" issue
# Keep only rows that have a 3-letter Code (these are the actual countries)
# This automatically removes 'Africa', 'Europe', etc., from the Entity column
df_countries = df[df['Code'].notna()].copy()

[ ] # 4. Remove "Impossible" values (Negatives or weird 3-digit outliers if they are errors)
# We assume CMR should be between 0.1 and 500 (standard range)
# And GDP must be positive
df_clean = df_countries[
    (df_countries['CMR'] > 0) &
    (df_countries['CMR'] < 500) &
    (df_countries['GDP'] > 0)
]
```

DATA PREPARATION

```
-- This query creates a unified view where every country now has its continent
SELECT
  m.Entity,
  m.Code,
  m.Year,
  m.CMR,
  m.GDP,
  c.region
FROM child_mortality_cleaned AS m
LEFT JOIN continents2 AS c ON m.Code = c.`alpha-3`
WHERE m.Code IS NOT NULL;

-- Add to new table
CREATE TABLE child_mortality_data AS
SELECT
  m.Entity,
  m.Code,
  m.Year,
  m.CMR,
  m.GDP,
  m.Population,
  c.region
FROM child_mortality_cleaned AS m
LEFT JOIN continents2 AS c ON m.Code = c.`alpha-3`
WHERE m.Code IS NOT NULL;
```

Transformation (SQL)

Performed a LEFT JOIN to map countries to regions and created a permanent TABLE for reporting.





STATISITICAL DIVE

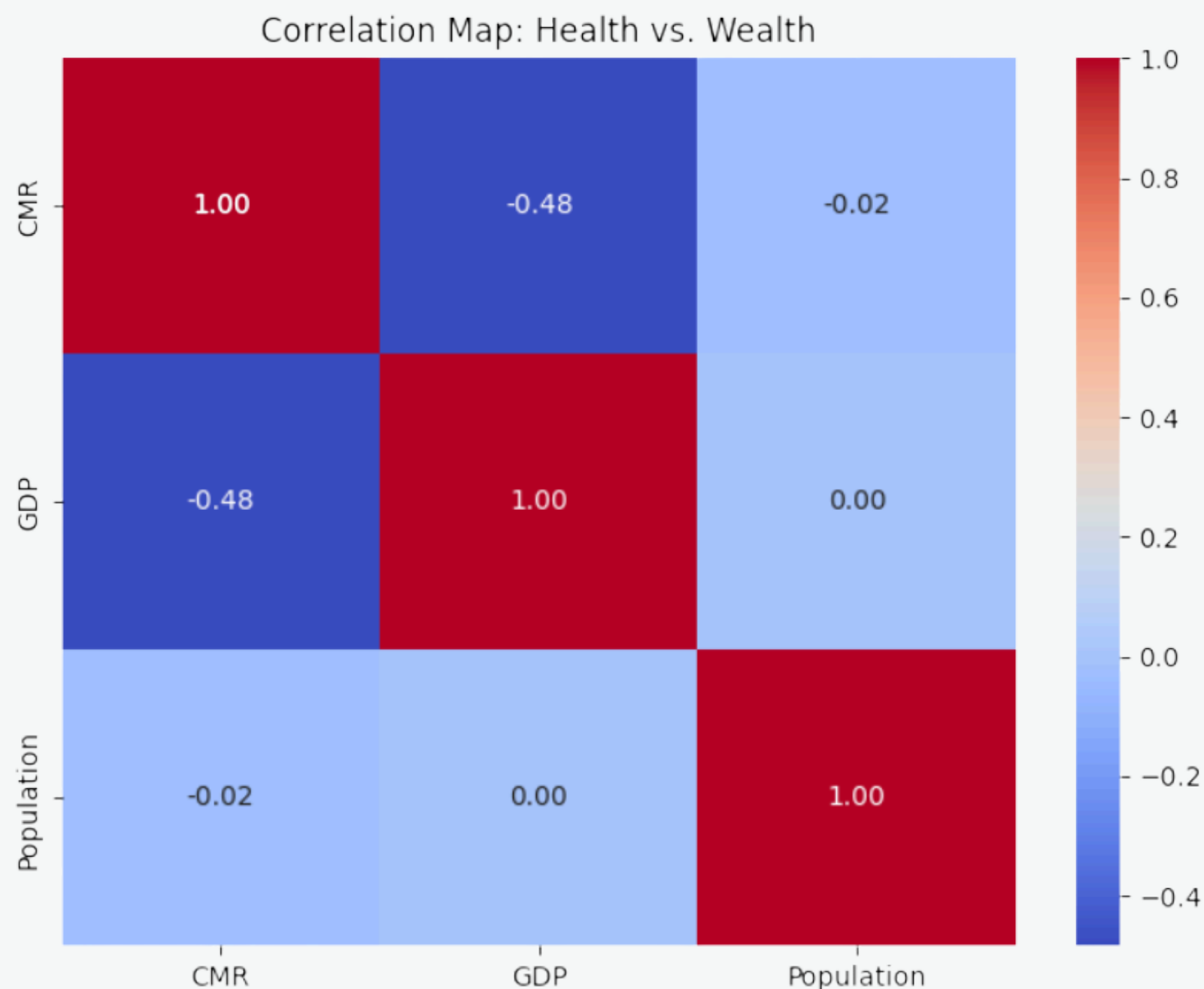
The Wealth-Health Link (CMR vs. GDP: -0.48)

As GDP per capita increases, the Child Mortality Rate (CMR) tends to decrease. This confirms the economic theory that wealthier nations generally have better healthcare infrastructure, nutrition, and sanitation.

The Population Factor (CMR vs. Population: -0.02)

The size of a country's population has almost no impact on its child mortality rate.

It proves that a country being "overpopulated" or "small" isn't a primary driver of how well its children survive; it's about the quality of the economic and health systems, not the quantity of people.



GLOBAL CHILD MORTALITY RATE DASHBOARD

Average Child Mortality Rate

10.74

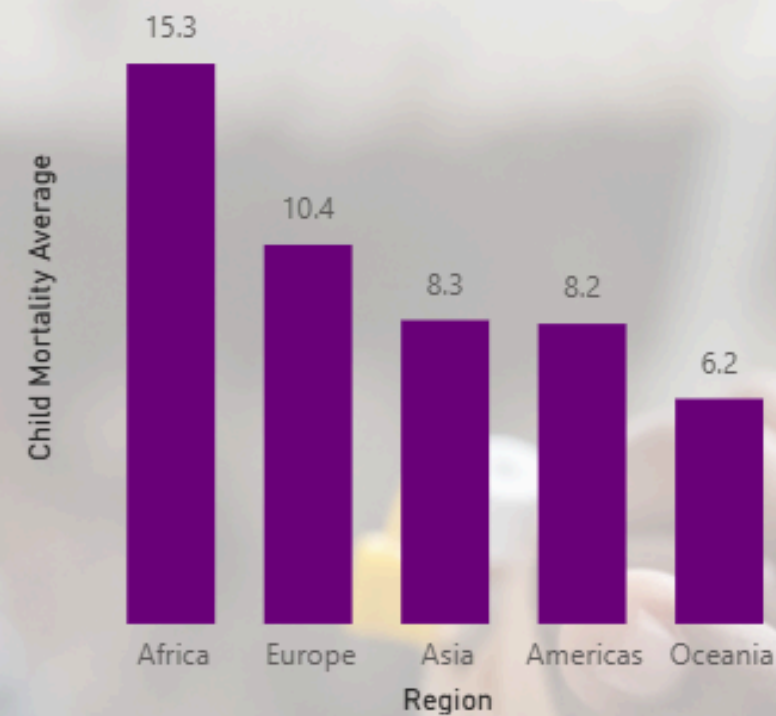
Total Population (2022)

8bn

Average Gross Domestic Product

9.64K

Regional Performance (Child Mortality Rate)

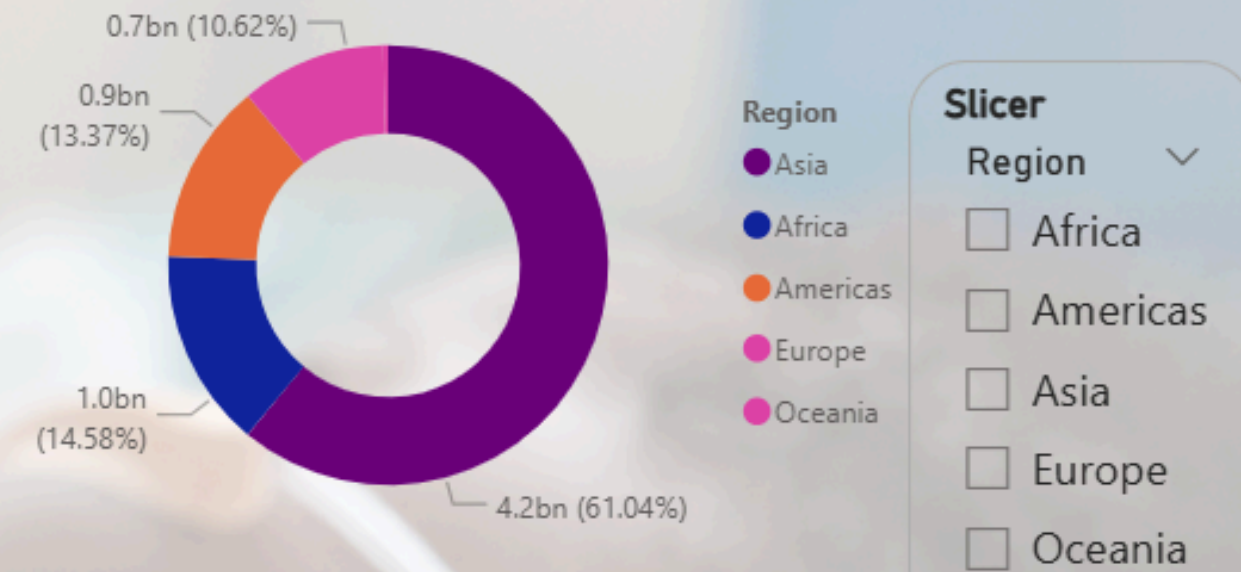


Correlation between GDP and Child Mortality Rate

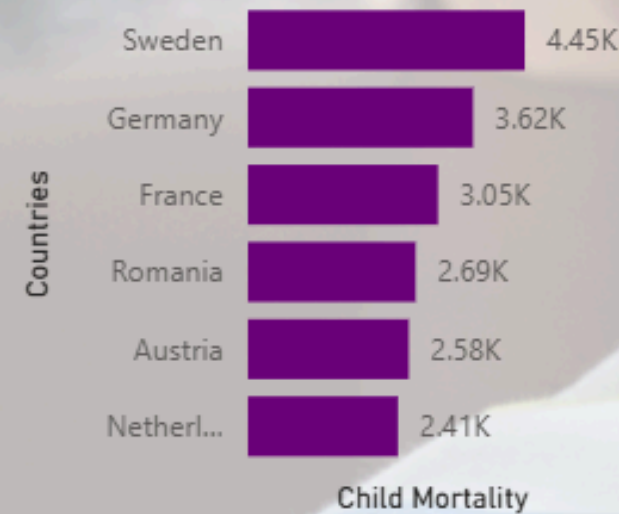
Entity: Countries



Population by Region



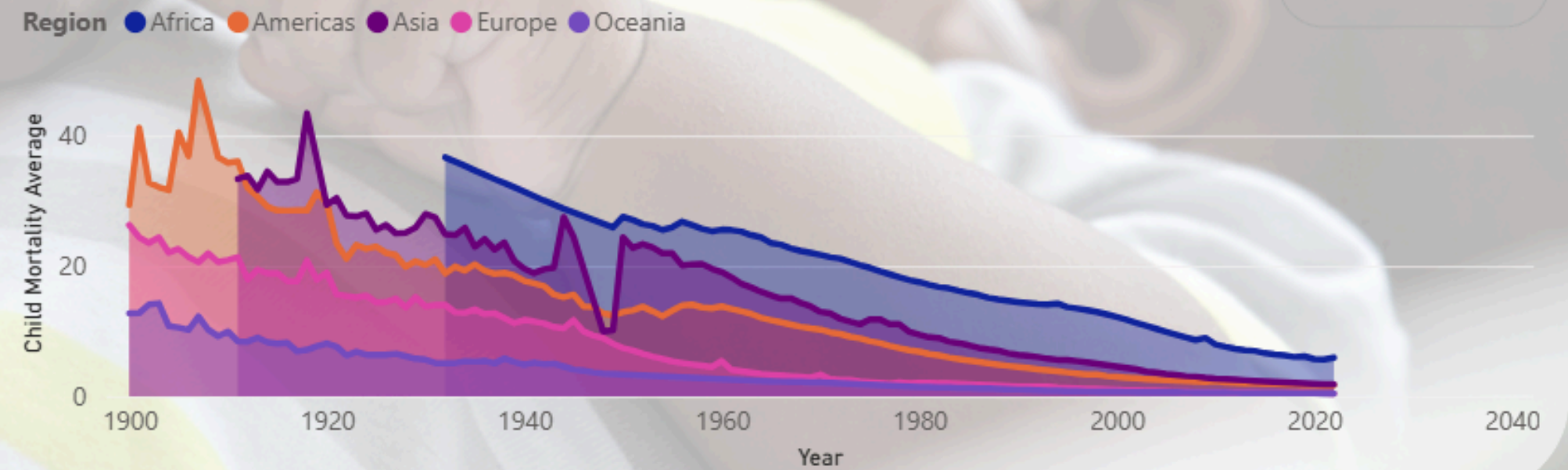
Top 6 Highest Mortality Entities



Top 6 Lowest Mortality Entities

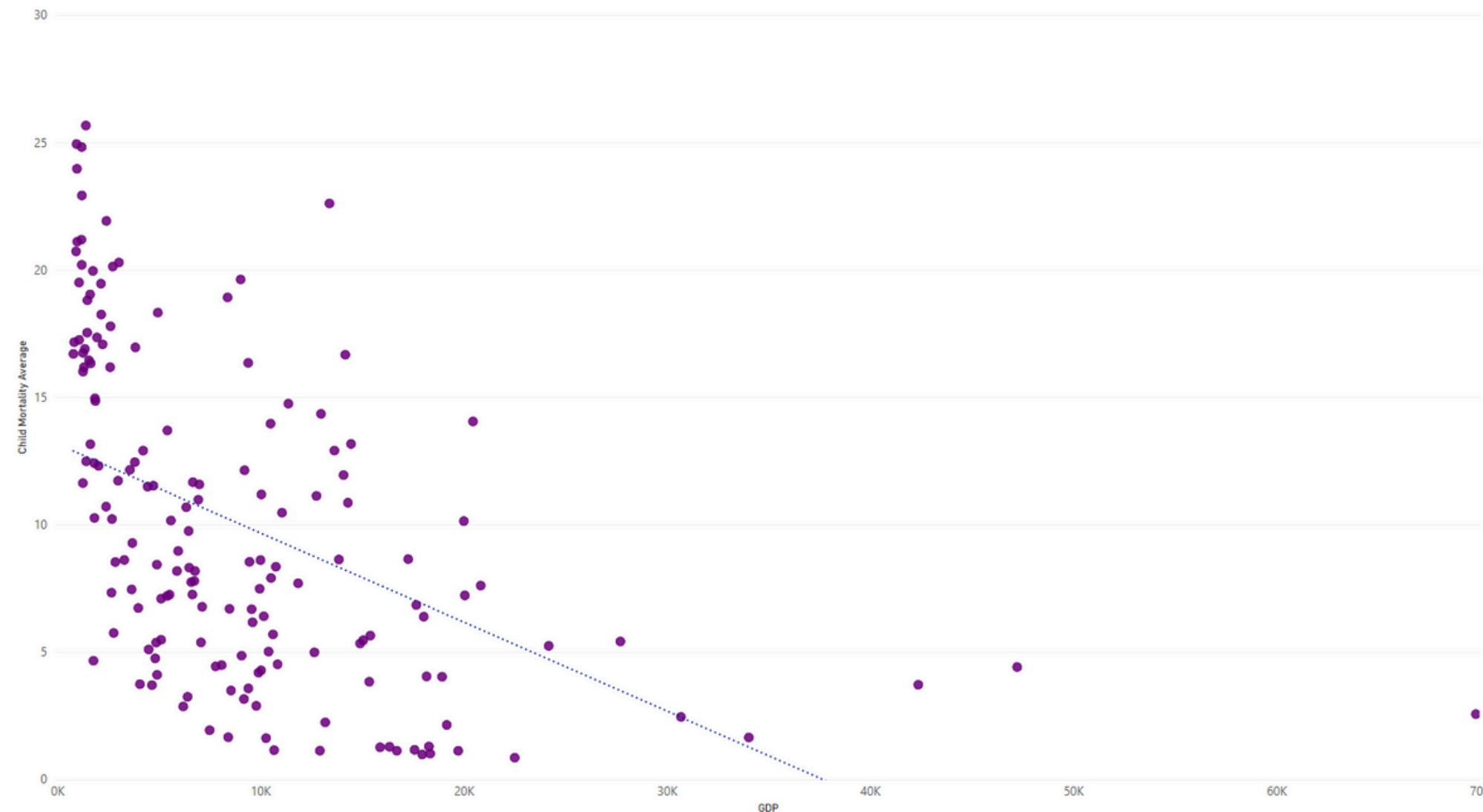


Average of CMR by Year and Region





INSIGHTS



Observation

Economic growth is a major driver of child survival, but it isn't the only one.

Evidence

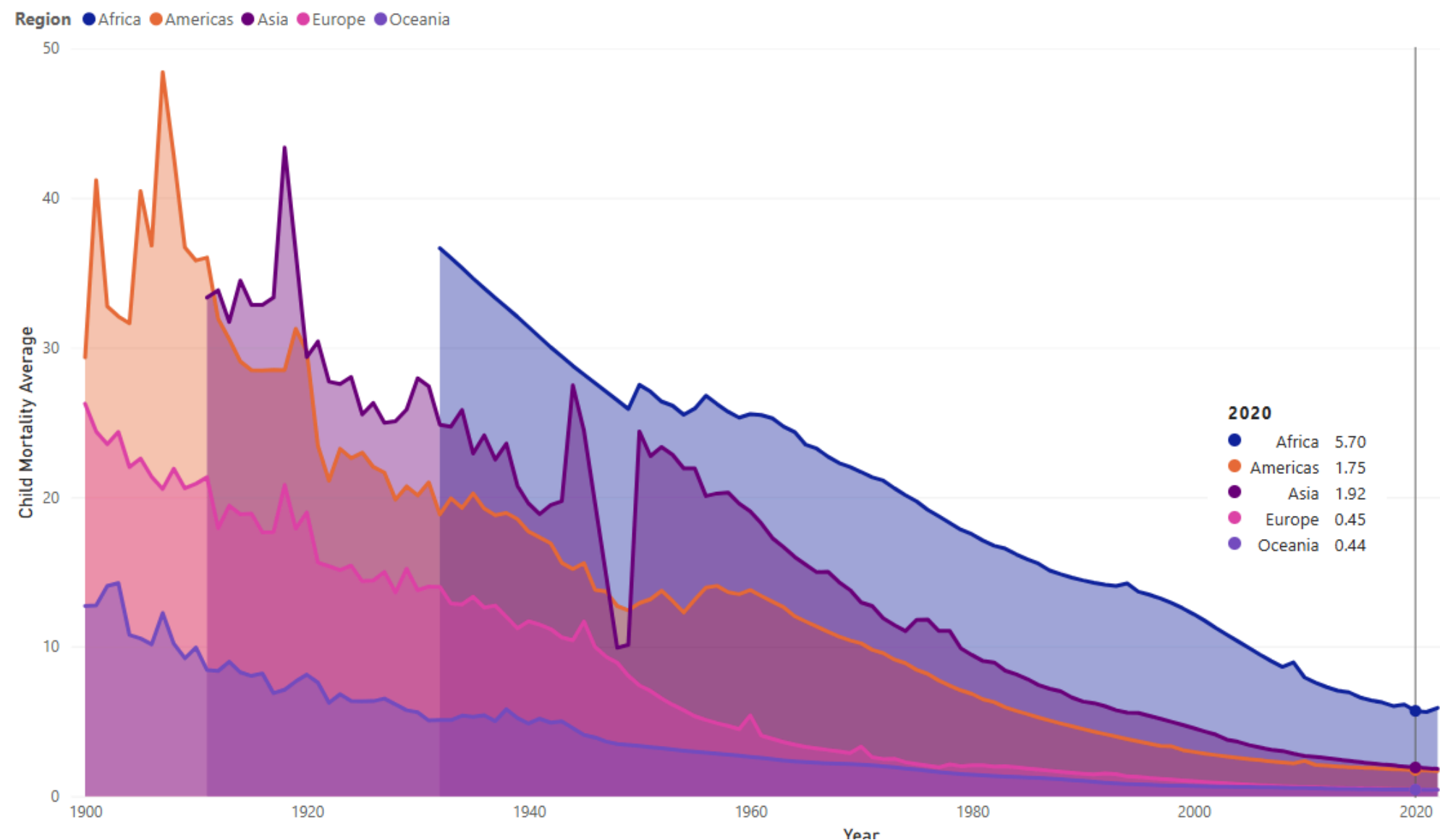
My Python analysis showed a **-0.48 correlation** between GDP and Child Mortality. In the Power BI scatter plot, we see 'High-Performing' outliers—countries with a GDP under \$5,000 achieving mortality rates lower than countries with double their wealth.

Conclusion

National wealth provides the *opportunity* for better health, but effective policy and healthcare distribution are what actually *deliver* results.



INSIGHTS



Observation

We are living in the most successful era for child survival in human history.

Evidence

The **Area Chart** shows a consistent downward trend across every single region since 1960. Even in regions currently facing crises, the current mortality rate is significantly lower than the global average was 40 years ago.

Conclusion

The 'Global Health Playbook' is working. The focus should now shift from starting progress to accelerating it in the 6 highest-mortality countries identified in our Bar Chart.



INSIGHTS

**Which countries have
"High Health Efficiency"?**

**(Low GDP but also Low
Mortality)**

««	Entity	▼	Code	▼	CMR	▼	GDP	▼	region	▼
	Filter	⋮ ▼	Filter	⋮ ▼	Filter	⋮ ▼	Filter	⋮ ▼	Filter	⋮ ▼
1	Cuba		CUB		0.79		7648.704		Americas	
2	Ukraine		UKR		0.85		7749.0996		Europe	
3	El Salvador		SLV		1.08		9218.905		Americas	
4	Cape Verde		CPV		1.21		7202.767		Africa	
5	Jordan		JOR		1.36		9151.479		Asia	
6	Nicaragua		NIC		1.39		5092.8413		Americas	
7	Palestine		PSE		1.42		4775.202		Asia	
8	Sao Tome and Principe		STP		1.46		3751.6428		Africa	
9	Moldova		MDA		1.48		6674.871		Europe	
10	Honduras		HND		1.6		5186.5303		Americas	



CONCLUSION

Summary

The project successfully mapped the 60-year decline of CMR and proved that while GDP helps, it is not "destiny".

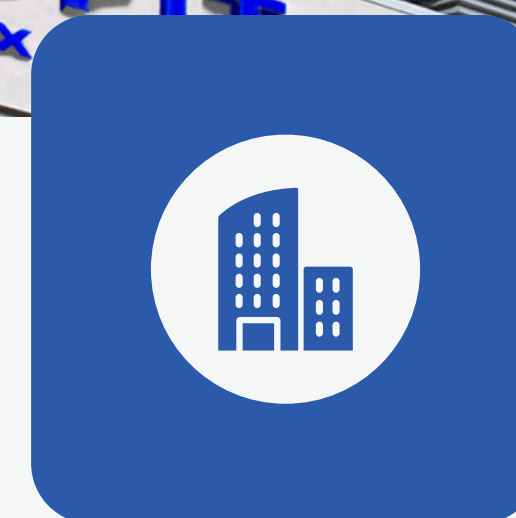
Next Steps

If given more time, I would join this data with 'Education Spending' or 'Healthcare Access' metrics to further explain the variance not covered by wealth alone.





HOSPITAL
Complex



THANK YOU!